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Featured Letter

Structure-Based Design and Evaluation of Reversible KRAS G13D Inhibitors

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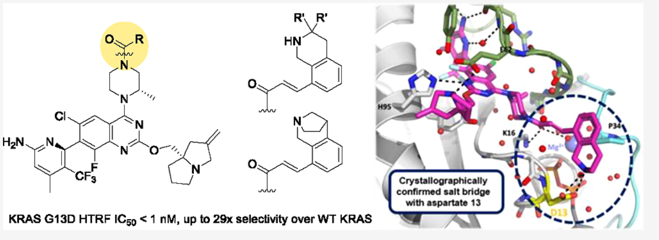
ABSTRACT: Oncogenic *KRAS* mutations were identified decades ago, yet the selective inhibition of specific KRAS mutant proteins represents an ongoing challenge. Recent progress has been made in targeting certain P-loop mutant proteins, in particular KRAS G12C, for which the covalent inhibition of the GDP state via the Switch II pocket is now a clinically validated strategy. Inhibition of other KRAS mutant proteins such as KRAS G13D, on the other hand, still requires clinical validation. The remoteness of the D13 residue relative to the Switch II pocket in combination with the solvent exposure and conformational flexibility of the D13 side chain, as well as the difficulties of targeting carboxylate residues covalently, renders this specific protein particularly challenging to target selectively. In this report, we describe the design and evaluation of potent and KRAS G13D-selective reversible inhibitors. Subnanomolar binding to the GDP state Switch II pocket and biochemical selectivity over WT KRAS are achieved by leveraging a salt bridge with D13.



KEYWORDS: *KRAS G13D, GTPases, Structure-based drug design, Switch II pocket, Oncology*

RAS, which encompasses the three isoforms KRAS, NRAS, and HRAS, constitutes a family of membrane-bound GTPases that function as molecular switches and regulate critical downstream signaling cascades via the GDP-bound "off" state and the GTP-bound "on" state.[1] KRAS in particular, which has been recognized as the most frequently mutated oncoprotein in cancer,[2] has received significant attention from researchers in academia and the pharmaceutical industry alike.[3] Long considered a challenging or even intractable target for small-molecule pharmaceuticals due to its featureless surface with an apparent lack of druggable pockets, KRAS moved back into the medicinal chemistry spotlight, in part, after a report by Shokat and co-workers in 2013 related to KRAS G12C.[4] Cysteine 12 (C12) in KRAS G12C is located on the P-loop (residues 10–15) in close proximity to both the nucleotide pocket and the two switch regions, i.e., switch I (SWI, residues 29–38) and switch II (SWII, residues 57–74), which are involved in critical effector interactions. Using cysteine-reactive molecular fragments, Shokat and co-workers discovered an allosteric pocket largely composed of the SWII loop region (and hence called the SWII pocket), thus opening an avenue toward targeting KRAS G12C.[4]

To date, two covalent KRAS G12C (GDP) targeting small-molecule inhibitors that bind to the SWII pocket have been approved by the FDA, namely, sotorasib (**1**) and adagrasib (**2**) (see Figure 1a),[5,6] while several others, such as divarasib (GDC-6036, **3**),[7,8] which emerged from our own KRAS G12C

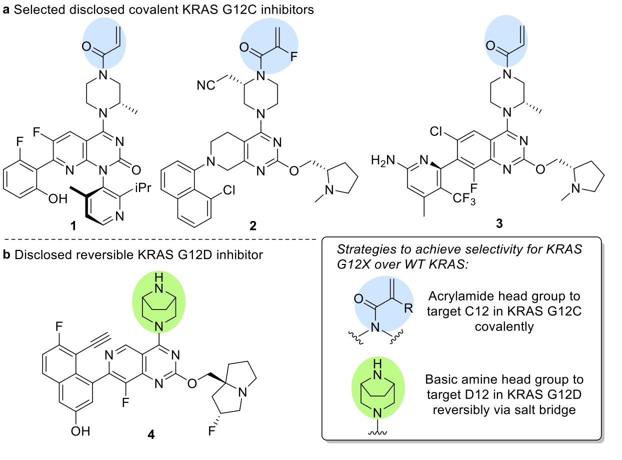


Figure 1. Structures of selected KRAS G12X (GDP) inhibitors that bind to the SWII pocket, highlighting the applied strategies to achieve selectivity over WT KRAS. (a) Approved KRAS G12C inhibitors sotorasib (**1**)[5] and adagrasib (**2**)[6] and clinical candidate divarasib (GDC-6036, **3**).[7,8] (b) Clinical candidate KRAS G12D inhibitor MRTX1133 (**4**).[11]

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program, and opnurasib (JDQ443), whose structure was recently disclosed by Novartis,[9,10] are undergoing clinical trials. Other KRAS mutant proteins devoid of an appropriately positioned nucleophilic cysteine for covalent modification (such as G12D, G12V and G13D) have proven more recalcitrant toward selective inhibition, and more limited progress has been reported so far. Recently, researchers at Mirati Therapeutics published their work that culminated in the discovery of a reversible KRAS-G12D-selective small-molecule GDP state inhibitor, MRTX1133 (**4**) (Figure 1b).[11] Notably, this molecule demonstrated a pharmacodynamic effect (pERK modulation) and efficacy in a Panc 04.03 model upon intraperitoneal (ip) administration and is now being evaluated in Phase I clinical studies.

KRAS G13D mutations occur in approximately 7−10% of colon cancer cases in the United States,[12,13] but the discovery of a selective small-molecule inhibitor of the KRAS G13D protein has been elusive so far. While seemingly closely related to other P-loop mutations such as G12D, the location of the G13D mutation at the 13 position presents several unique challenges. Most notably, these include the remoteness of aspartate 13 (D13) from the SWII pocket, the expected conformational flexibility of the aspartate side chain, and the solvent exposure, which suggests that a desolvation penalty might offset any potential affinity gain of a salt bridge with D13.[14] The latter appears particularly disconcerting, as the formation of a salt bridge with D13 would appear to be the most direct strategy toward achieving KRAS G13D selectivity over WT KRAS. Furthermore, biochemical and biological challenges associated with targeting KRAS G13D include the higher nucleotide exchange rate and lower nucleotide affinity[15] relative to other KRAS mutants, raising some important questions about the feasibility of targeting the GDP-bound state of KRAS G13D. To assess these anticipated challenges, we initiated a research project aimed at evaluating the feasibility of targeting the GDP state of KRAS G13D potently and, importantly, selectively over WT KRAS. Herein we report initial results of this endeavor.

The chemical matter derived from research efforts toward the discovery of covalent KRAS G12C inhibitors was considered a good starting point for further optimization. In particular, in the context of our KRAS G12C program we had explored the oxazepino−quinazoline scaffold as present in **5** (Figure 2), which caught our attention early on, as it suggested the advantage of maximal conformational restriction of the piperazine acrylamide headgroup.[16,17] An evaluation of **5** in paired KRAS G13D and WT KRAS TR-FRET assays using recombinant proteins to monitor inhibition of GDP exchange revealed a biochemical IC_{50} of 0.63 μM for KRAS G13D with no significant selectivity (i.e., 1.6×) for KRAS G13D over WT KRAS, as expected.

The corresponding X-ray structure (Figure 2) revealed a binding pose consistent with previously described SWII pocket binders,[6,11] with the piperazine acrylamide headgroup extending out of the SWII pocket into a shallow channel between the SWI loop and the P-loop. Intriguingly, the structure showed the D13 side chain (highlighted in yellow) in a conformation that points *away* from the SWII pocket that is occupied by the ligand (termed the "D13-out" conformation). Although this conformational preference, if confirmed, would present a formidable challenge for targeting D13 from the SWII pocket, we reasoned that the observed D13-out conformation was likely stabilized by crystal packing contacts

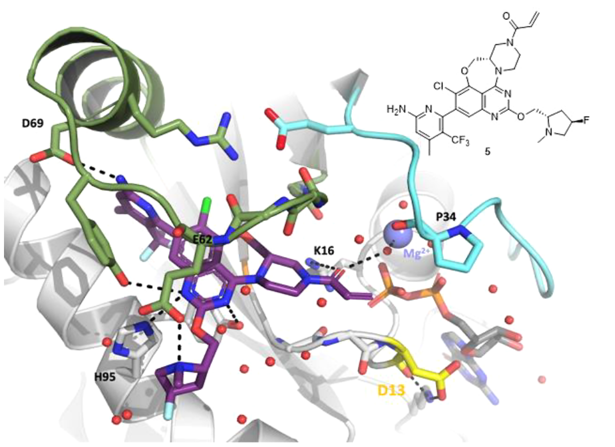

Figure 2. Crystal structure of KRAS G13D-GDP in complex with **5** (2.1 Å; PDB code 8UN3). SWI residues are colored cyan, SWII residues are colored green, and main-chain atoms of the P-loop are shown as sticks with D13 colored yellow.

with residues from a noncrystallographic symmetry-related molecule. Nonetheless, achieving a better understanding of the conformational preferences of the D13 side chain in KRAS G13D was deemed critical. Insights into the conformational preferences of the D13 side chain were obtained by molecular dynamics simulations of the KRAS G13D-GDP complex, indicating that a conformation in which the D13 side chain pointed *toward* the SWII pocket (termed the "D13-in" conformation) was highly populated. Importantly, this computational result provided us with more confidence in our approach that would rely on achieving biochemical selectivity for KRAS G13D over WT KRAS through a salt bridge with D13, and we hypothesized that the D13-in conformer would ultimately be a viable conformer to target.

Our efforts began with a virtual screen using commercially available amino-substituted carboxylic acids and the cocrystal structure of **5** with KRAS G13D, hypothesizing that the piperazine moiety of **5** might provide a rigid framework for the attachment of a basic amine via an amide linkage. A focused set of representative amide analogs to systematically interrogate the positioning of the amine relative to D13 were synthesized and biochemically evaluated. Unfortunately, none of the initially profiled screening sets displayed meaningful selectivity (Table 1), providing an early hint at the challenges associated with our strategy of achieving mutant selectivity via a salt bridge with D13. It is also worth noting that these compounds generally showed either reduced or in some cases similar potency relative to compound **5**, suggesting that no additional beneficial interactions were created (based on available data, replacement of the fluoropyrolidine as present in **5** with a non-fluorinated prolinol as present in **7**−**26** was not expected to have a significant impact on biochemical potency). However, despite this initially disappointing outcome, these results suggested that the anticipated conformational flexibility of this series of amide analogs would hinder an effective salt bridge interaction with D13. Therefore, we decided to focus future structure-based design efforts on analogs with a greater degree of conformational restriction.

The cocrystal structure of **5** and KRAS G13D (GDP) (Figure 2) suggested that β-substituted *trans*-acrylamides could provide a suitable vector for the targeted D13-in conformer. Accordingly, an initial set of disubstituted acrylamides were

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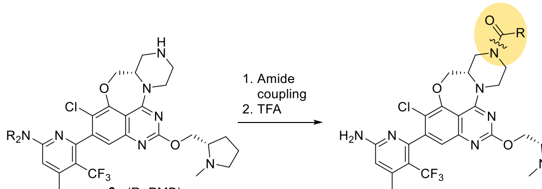
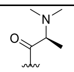
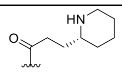
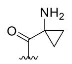
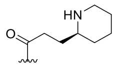
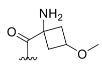
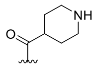
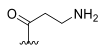
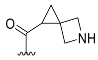
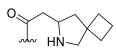
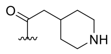
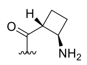
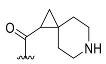
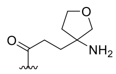
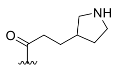
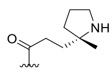
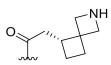
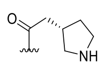
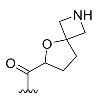
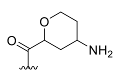
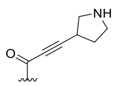
Table 1. General Synthesis and Biochemical Evaluation of a Focused Oxazepino−Quinazoline Library

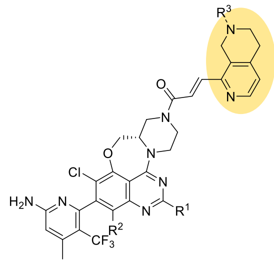


6a (R=PMB)
6b (R=H)
7-26

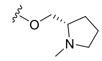
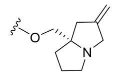
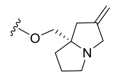
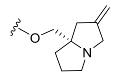
Cmpd.	Amide	G13D GDP IC$_{50}$/μM (Sel.)[a]	Cmpd.	Amide	G13D GDP IC$_{50}$/μM (Sel.)[a]
7		8.2 (0.94x)	17		2.0 (2.3x)
8		0.55 (1.8x)	18		1.7 (2.5x)
9[b]		0.88 (1.5x)	19		3.2 (2.2x)
10		2.7 (2.5x)	20[b]		2.4 (1.3x)
11[b]		9.0 (1.1x)	21		4.8 (2.0x)
12		3.0 (1.5x)	22[b]		2.4 (2.1x)
13[b]		1.3 (2.5x)	23[b]		2.1 (2.0x)
14		1.9 (2.4x)	24		3.5 (1.1x)
15		2.6 (2.1x)	25[b]		1.6 (2.5x)
16[b]		14 (0.41x)	26[b]		1.6 (2.3x)

[a]KRAS G13D GDP HTRF IC$_{50}$ (in μM) and IC$_{50}$ selectivity over WT KRAS (mean of n = 2). See the Supporting Information for assay conditions. [b]Mixture of diastereomers.

Table 2. Initial Biochemical Evaluation of Moderately KRAS G13D-Selective Tetrahydronaphthyridyl (THN) Acrylamides in the Oxazepino Series (27−31)



27-31

Cmpd.	R^1	R^2	R^3	KRAS G13D GDP IC$_{50}$/μM (Sel. over WT)[a]
27		H	H	0.18 (6.1x)
28		H	H	0.013 (7.3x)
29		F	H	0.0015 (6.7x)
30	H	F	H	2.7 (5.9x)
31		H	Me	0.049 (0.88x)

[a]KRAS G13D GDP HTRF IC$_{50}$ (in μM) and IC$_{50}$ selectivity over WT KRAS (mean of n = 2). See the Supporting Information for assay conditions.

designed, synthesized and evaluated. Indeed, the discovery of acrylamide **27**, the first example of a compound with moderate biochemical selectivity, albeit with only weak potency (0.18 μM), suggested that this strategy might prove more fruitful (Table 2). While modeling suggested the moderate selectivity to stem from an interaction of the protonated tetrahydronaphthyridine of **27** with D13, initial challenges related to obtaining a KRAS G13D cocrystal structure with **27** hampered validation of this hypothesis. **27** features a rigid tetrahydronaphthyridyl (THN) headgroup, in which the pyridine nitrogen was originally hypothesized to impart a coplanar arrangement of the pyridine and olefin moieties. The potency of **27** was further improved by about 14-fold through replacement of the prolinol C2 substituent with a methylene pyrrolizidine,[18] which led to **28**. Attachment of the 8-F substituent (to lock the desired aminopyridine atropisomer in place)[7] gave **29**, which displayed a biochemical IC$_{50}$ of 1.5 nM and comparable moderate KRAS G13D selectivity of 6.7×. Synthesis of compound **30** (R^2 = H), which is about 1800-fold less potent than **29**, demonstrated that a basic amine C2 substituent to engage glutamate 62 (E62) would be necessary to achieve nanomolar or potentially subnanomolar potency within this series. Finally, it is worth noting that tertiary amine **31** lost the moderate KRAS G13D selectivity observed for its secondary amine counterpart **28**.

The hypothesized advantage of an oxazepine tether (*i.e.*, conformational rigidity) came at the cost of increased synthetic complexity, higher TPSA, and slightly higher MW compared to the corresponding untethered scaffold as present in divarasib (**3**) (Figure 1a). With the discovery of an initial path toward selectivity, it was thus decided to interrogate this underlying hypothesis in more detail. Interestingly, transfer of the THN acrylamide moiety to the corresponding "untethered" methylpiperazine scaffold (**29** vs **32**; Figure 3) revealed a similar

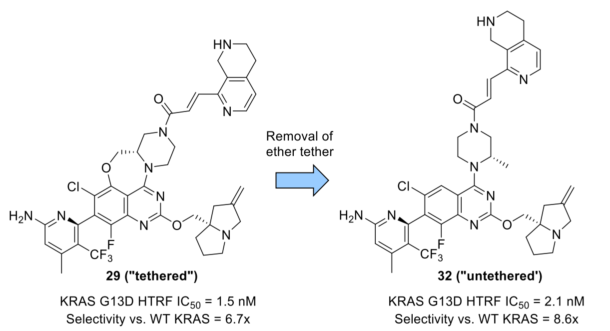


Figure 3. Molecular structures and biochemical potencies and KRAS G13D selectivities of **29** and **32**.

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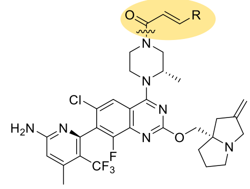
biochemical potency range for both series. The fact that the biochemical selectivity of **32** was similar to that of **29** (if not slightly improved) highlighted that removal of the oxazepine tether was feasible. This simplified our synthetic efforts going forward and also promised certain potential property space advantages as outlined above (*e.g.*, TPSA). Thus, "untethered" analog **32** was used as a starting point for further C4-substituent headgroup optimization (Table 3).

The focus of further design was directed toward optimizing the amine positioning to maximize the affinity gain from the critical salt bridge with D13.[19] Starting from **32**, initial headgroup deconstruction and amine repositioning led to the discovery of benzylic amine **33** (Table 3), which showed subnanomolar potency but somewhat reduced biochemical selectivity (IC_{50} = 0.53 nM, 5.7× selective) compared to **32**. Nonetheless, **33** was considered interesting, as it promised to enable the exploration of slightly different vectors to the more rigid THN headgroup series. Notably, the corresponding morpholine **34** did not show any significant KRAS G13D selectivity, which was not unexpected given the significantly reduced pK_a of the amine (measured pK_a for **34** = 5.4; measured pK_a for **33** = 8.4) and pointed at the inability of **34** to effectively form a salt bridge with D13 as a result of the lack of sufficient protonation at physiological pH. Installation of a methyl group *ortho* to the benzylic amine in **33** was expected to reduce the conformational flexibility of this headgroup; gratifyingly, the corresponding analog **35** indeed showed improved selectivity (9.6×). Tertiary amine **36** maintained potency and selectivity relative to its secondary amine counterpart **35**, and a KRAS G13D cocrystal structure with the former analog was obtained (Figure 4).

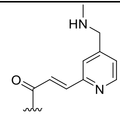
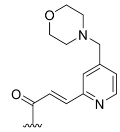
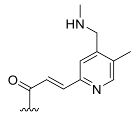
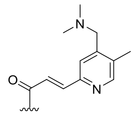
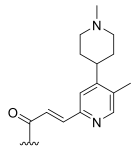
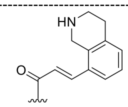
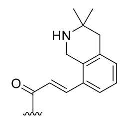
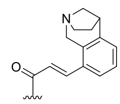
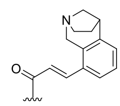
Overall, the crystallographically observed binding mode of **36** was similar to that of the previous structures and consistent with the expected binding mode based on modeling. Importantly, the crystal structure revealed that D13 adopts the D13-in conformation, in good agreement with the molecular dynamics simulations. The structure further showed an edge-to-face van der Waals contact of the pyridine moiety with the proline 34 (P34) side chain, while the tertiary amine was positioned within hydrogen-bonding distance to D13 (distance between tertiary amine and closest carboxylate oxygen ≈ 2.8 Å). Based on a comparison with the nonselective inhibitors in Table 1, we hypothesized that this combination of amine placement and the outlined hydrophobic interaction contributed to the selectivity of **36**, which served as a basis and rationale for further designs. In particular, optimization of the amine positioning and further rigidification led to **37**, showing a subnanomolar biochemical IC_{50} of 0.34 nM combined with 15× selectivity over the WT, the highest selectivity yet observed.

Unfortunately, further attempts to improve the biochemical selectivity in this subseries were unsuccessful, prompting us to reconsider **32** as a starting point to optimize the selectivity. Along this line, the necessity of a pyridine nitrogen was probed by replacing the THN with a tetrahydroisoquinoline (THIQ) headgroup (Table 3). This modification provided **38**, which featured slightly increased potency compared to **32** (*ca.* 3-fold) and a moderate increase in KRAS G13D selectivity. This compound was successfully cocrystallized with KRAS G13D (GDP). The resulting structure features the expected salt bridge with D13, with the D13 side chain again adopting the D13-in conformation (Figure 5).

Table 3. Biochemical Potency/Selectivity of "Untethered" Piperazine Acrylamides 33−41



33-41

Cmpd.	Amide	KRAS G13D GDP IC_{50}/nM (Selectivity over WT)[a]
33		0.53 (5.7x)
34		4.1 (1.2x)
35		0.50 (9.6x)
36		0.52 (9.2x)
37		0.34 (15x)
38		0.69 (11x)
39		0.46 (16x)
40[b]		0.55 (27x)
41[b]		0.41 (29x)

[a]KRAS G13D GDP HTRF IC_{50} (in nM) and IC_{50} selectivity over WT KRAS (mean of n = 2). See the Supporting Information for assay conditions. [b]Single diastereomer with unknown absolute configuration.

The carboxylate oxygen atoms of D13-in are equidistant (3.1 Å) from the THIQ secondary amine, which is expected to be protonated. The THIQ moiety is positioned orthogonal to the acrylamide linker and packs up against the P34 side chain, while the acrylamide carbonyl retains the hydrogen-bonding interaction with the lysine 16 (K16) side chain and the water-

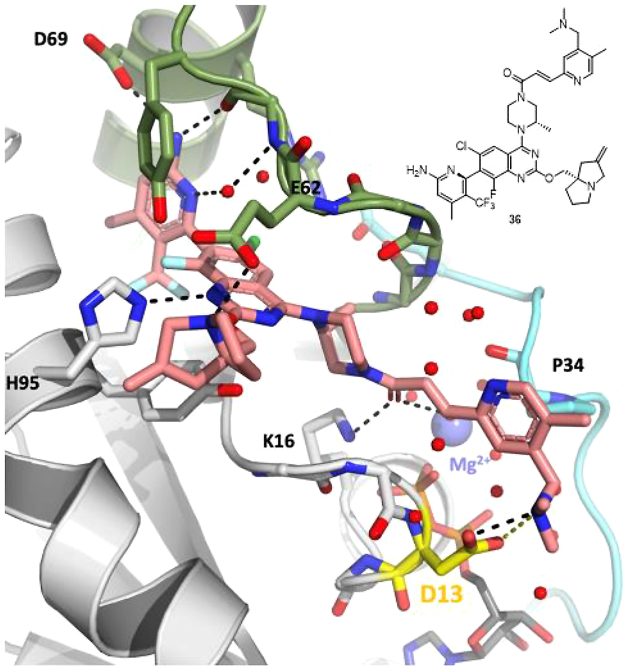


Figure 4. Cocrystal structure of KRAS G13D-GDP in complex with **36** rendered in salmon (1.6 Å; PDB code 8UN4). Coloring and representation are according to Figure 2, with D13 highlighted in yellow.

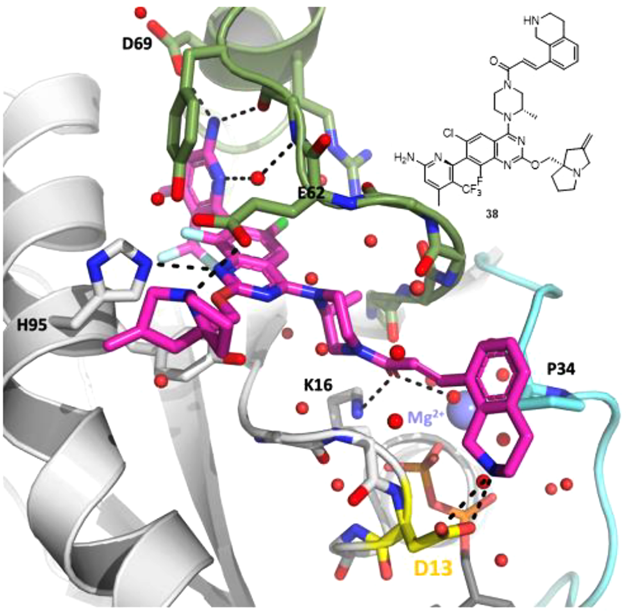


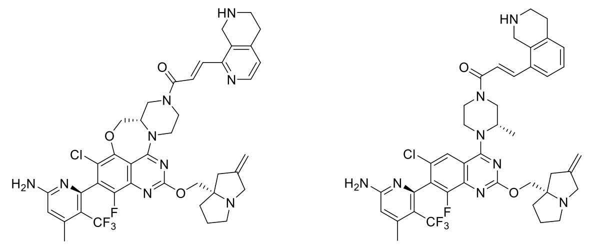
Figure 5. Crystal structure of KRAS G13D-GDP in complex with **38** (1.3 Å; PDB code 8UN5). Coloring and representation are according to Figure 2, with D13 highlighted in yellow.

mediated interaction with Mg^{2+}. The interactions in the SWII pocket residues were unchanged.

Installation of a *gem*-dimethyl group α to the basic nitrogen (*i.e.,* **39**) further increased the selectivity to 16×, potentially as a result of shielding of the salt bridge with D13 from solvent.[20] Diastereomers **40** and **41** were then prepared with the intention to further restrict the conformation of the basic amine and replace the secondary amine with a tertiary amine, which was regarded as beneficial for later optimization efforts in light of the reduced hydrogen-bond-donor count and expected permeability advantages. Despite the discouraging result that had been previously observed for tertiary amine **31**, modeling suggested that a "tied-back" tertiary amine could be tolerated and capable of forming a productive salt bridge with D13 as in its protonated form, and the proton would be expected to point toward the carboxylate of D13. Indeed, the KRAS G13D selectivities of both diastereomers **40** and **41** proved promising (27× and 29×, respectively), rendering these two compounds the most selective KRAS G13D inhibitors described to date. Thus, **40** and **41** support the feasibility of inhibiting KRAS G13D selectively over WT KRAS in the GDP state with small-molecule SWII binders.[21]

Representative compounds in the oxazepino and untethered quinazoline series (i.e., **29** and **38**, respectively) were further benchmarked *in vitro* and *in vivo* (Table 4). Both compounds showed only limited antiproliferative activity in the KRAS G13D cell line HCT116 (7 day assay). It is noteworthy that in general, compounds within these series displayed rather low permeability in MDCK cells (likely as a result of the dibasic nature of these molecules), which possibly contributes to the high cell shift. In a mouse iv PK experiment at a dose of 1 mg/kg, both compounds **29** and **38** showed low clearance (*ca.* 3–4% of mouse liver blood flow). The corresponding volume of distribution at steady-state (V_{ss}) was approximately 60% (0.42 L/kg) to 120% (0.9 L/kg) of total body water volume for **29** and **38**, respectively. This resulted in terminal half-lives of 3.2 h for **29** and 7.6 h for **38**.

Table 4. *In Vitro* (HTRF and CTG Potency) and *In Vivo* (Mouse Pharmacokinetics)[a] Profiles of 29 and 38



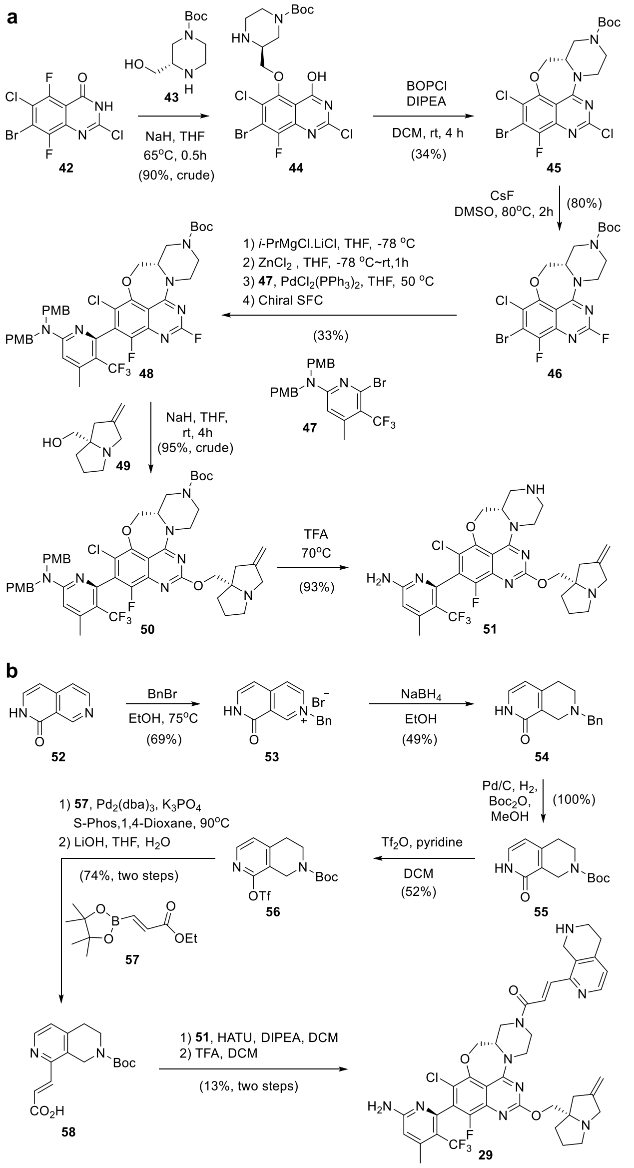
29 38

	29	38
KRAS G13D GDP HTRF IC_{50} (sel. over WT)	1.5 nM (6.7×)	0.69 nM (11×)
HCT116 (G13D) IC_{50}	6.1 μM	2.1 μM
CL	2.9	3.4
V_{ss}	0.42	0.9
$t_{1/2}$	3.2	7.6

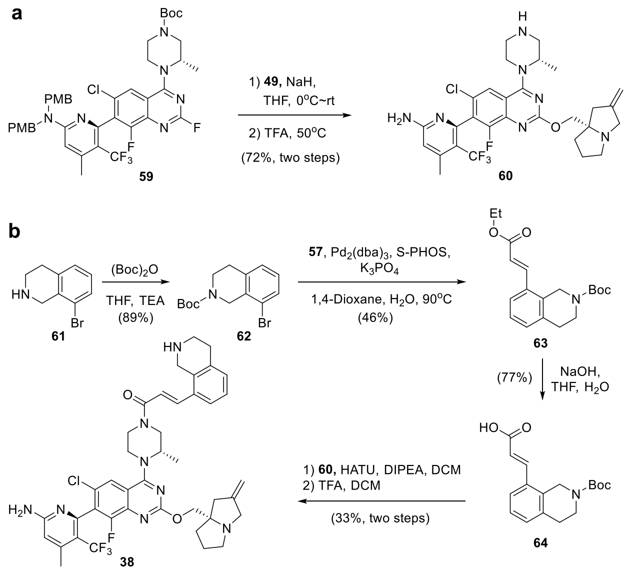
[a]Compounds were dosed at 1 mg/kg (iv) in 10% DMSO, 35% PEG400, 55% water. Clearance (CL) is in mL min^{-1} kg^{-1}; volume of distribution at steady state (V_{ss}) is in L/kg; half-life ($t_{1/2}$) is in h.

Two representative syntheses are shown in Scheme 1 and Scheme 2. The synthesis of **29** commenced with quinazolinone **42**,[18] which was subjected to an S_NAr reaction with alcohol **43** to give **44** (see Scheme 1a). BOPCl-mediated cyclization gave **45**, which was then reacted with CsF at elevated temperature to replace the 2-Cl substituent with fluoride (see **46**) in order to facilitate an S_NAr reaction later on. Negishi coupling with **47**[8] followed by chiral separation of the resulting atropisomers gave **48**. The 2-F in **48** was displaced with amino alcohol **49**[18] to give **50**, and subsequent global deprotection with TFA delivered key intermediate **51**. The headgroup intermediate

Scheme 1. Synthesis of 29: (a) Synthesis of Oxazepino–Quinazoline Key Intermediate 51; (b) Synthesis of Headgroup Intermediate 58 and Completion of the Synthesis of 29



Scheme 2. Synthesis of 38: (a) Synthesis of Quinazoline Key Intermediate 60; (b) Synthesis of Headgroup Intermediate 64 and Completion of the Synthesis of 38



was synthesized as shown in Scheme 1b. Naphthyridinone **52** was *N*-benzylated (see **53**) and subsequently subjected to a $NaBH_4$-mediated reduction to give semisaturated intermediate **54**. Replacement of the Bn with a Boc protecting group to give **55** was followed by *O*-triflation to give **56** and Suzuki coupling with **57**. Subsequent ester saponification then provided THN acrylate **58**, which was coupled with **51** (HATU, DIPEA). Final deprotection afforded **29**.

The synthesis of **38** began with quinazoline **59**[8] (see Scheme 2a), which was subjected to an S_NAr reaction with **49** and PMB deprotection (TFA, 50 °C) to give **60**. The THIQ acrylamide moiety was synthesized from tetrahydroisoquinoline **61** in three steps, including Boc protection to give **62**, Suzuki coupling with **57** to give **63**, and ester hydrolysis to give **64** (see Scheme 2b). HATU-mediated coupling of the latter intermediate with **60** followed by deprotection provided **38**.

In conclusion, we herein describe the results of our feasibility analysis related to achieving high reversible affinity and biochemical selectivity for KRAS G13D over WT KRAS and disclose a novel class of KRAS G13D inhibitors that target the GDP or "off" state of KRAS G13D. Structure-based design-guided optimization of an initial hit (*i.e.*, **27**: IC_{50} = 0.18 μM, 6.1× biochemically selective over WT KRAS) delivered G13D-selective piperazine acrylamides with subnanomolar potency, such as **41** (IC_{50} = 0.41 nM, 29× biochemically selective over WT KRAS). To the best of our knowledge, these compounds represent the first published examples of biochemically selective KRAS G13D inhibitors and as such support the feasibility of selectively inhibiting KRAS G13D over WT KRAS. Our studies also highlight the challenges associated with targeting KRAS G13D with reversible small-molecule SWII binders. Most notably, these include achieving (1) high selectivity and (2) high potency while maintaining drug-like properties. These challenges will have to be addressed in any future research directed toward the design and identification of KRAS G13D inhibitors with potential applications in cancer therapy.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsmedchemlett.3c00478.

Experimental procedures for analog synthesis and characterization data; NMR spectra and LCMS chromatograms for compounds **27**–**41**; experimental procedures for X-ray crystallography and crystallographic data of **5**, **36**, and **38**; HTRF and cell viability assay conditions; experimental procedures for *in vivo* pharmacokinetics studies (PDF)

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Notes

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ABBREVIATIONS

BnBr, benzyl bromide; Boc, *tert*-butoxycarbonyl; Boc_2O, di-*tert*-butyl dicarbonate; BOPCl, bis(2-oxo-3-oxazolidinyl)-phosphinic chloride; CL, clearance; dba, *trans*,*trans*-dibenzylideneacetone; DCM, dichloromethane; DIPEA, diisopropylethylamine; DMSO, dimethyl sulfoxide; EtOH, ethanol; GDP, guanosine diphosphate; GSH, glutathione; GTP, guanosine triphosphate; HATU, 1-[bis(dimethylamino)methylene]-1*H*-1,2,3-triazolo[4,5-*b*]pyridinium-3-oxide hexafluorophosphate; HRAS, Harvey rat sarcoma virus oncogene; HTRF, homogeneous time-resolved fluorescence; ip, intraperitoneal; iPr, isopropyl; iv, intravenous; KRAS, Kirsten rat sarcoma virus oncogene; MeOH, methanol; mL, milliliter; NRAS, neuroblastoma RAS viral oncogene; pERK, phospho-ERK; PK, pharmacokinetics; PMB, *p*-methoxybenzyl; RAS, rat sarcoma virus oncogene; PEG, poly(ethylene glycol); rt, room temperature; SFC, supercritical fluid chromatography; $t_{1/2}$, half-life; TEA, triethyl amine; Tf2O, triflic anhydride; TFA, trifluoroacetic acid; THF, tetrahydrofuran; TPSA, topological polar surface area; V_{ss}, volume of distribution at steady state; WT, wild-type.

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(21) Given the presence of an acrylamide moiety and the potential risk of covalent adduct formation (nucleophilic amino acid residues, GSH, or other nucleophiles), selected compounds were subjected to a cysteine reactivity assay and were mostly found to be rather unreactive, likely due to steric and electronic deactivation of the acrylamide by the β-aryl substituent. Half-lives of representative compounds in a cysteine reactivity assay (1 μM test compound in a phosphate buffer solution containing 5 mM cysteine in the presence of 100 mM KPi buffer at pH 7.4): $t_{1/2}$(**35**) > 540 min; $t_{1/2}$(**36**) > 540 min; $t_{1/2}$(**38**) = 417 min.